EXPERIMENT - 2 DATE: 30/07/26

AIM: Study of different type of network. cables.

i) UNSHEILDED TWISTED PAIR (UTP) CABLE:

* Category 3:

Max Data Transmission: 10 bps.

Application: 10Base-T Ethernet

* Category 5:

Max Data Transmission: Upto 100 Mbps

Application: Fast Ethernet, GigaEthernet

* Category 5e:

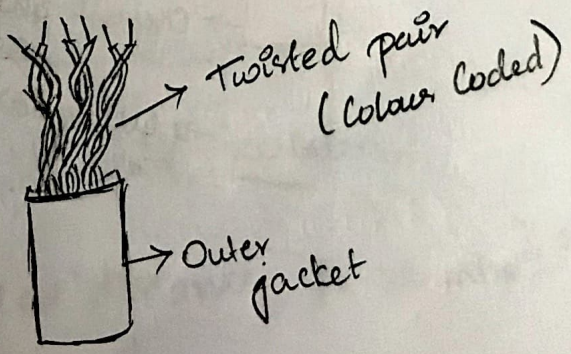
Application: Fast Ethernet, Gigabit Ethernet

Max Data Transmission: 1 Gbps.

Advantages: • Cheaper
• Easier installation

Disadvantage: More prone to EMI

Diagram:



ii) SHEILDED TWISTED PAIR (STP) CABLE:

* Category 6, 6a:

Max Data Transmission: 10 Gbps

Application: 10G Ethernet

* Category 7:

Max Data Transmission: 10Gbps

Application: 10G Ethernet (100)

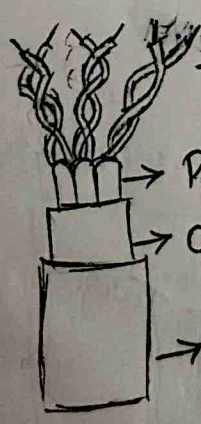
Advantage: • Sheilded

• Faster than UTP

Disadvantage: • Expensive

• Installation Difficult.

Diagram:



iii) COAXIAL CABLE:

* RG-6
  RG-59
  RG-11

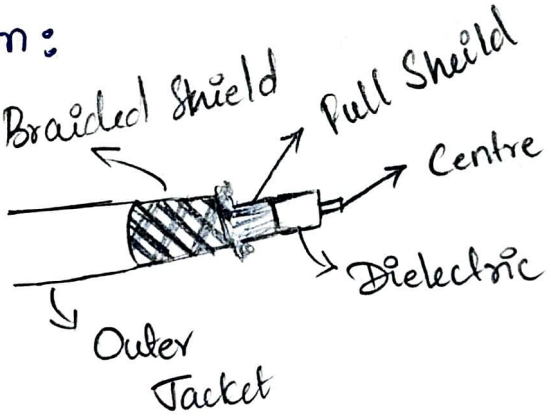
Max Data Transmission: 10 - 100 Mbps

Application: Speed of signal 500 m

Advantages:
- High Bandwidth
- Versatile
- Immune to interfaces

Disadvantages:
- Limited distance
- Bulky
- Cost High

Diagram:



iv) FIBRE OPTICS CABLE:

* SINGLE MODE
  MULTI MODE

Max Data Transmission: 100 Gbps

Application: Max distance is 100 mtrs.

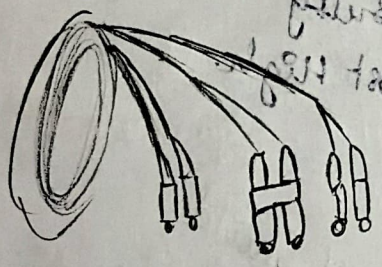
Advantage:

- High Speed
- High Security

Disadvantage:

- Expensive
- Required skills

Diagram:



RESULT: Successfully analysed the study of different types of network cables.